

HUTCHINSON, KS, USA

WMO: 724506

Lat: 38.065N

Lon: 97.861W

Elev: 470

StdP: 95.80

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13986

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-12.1	-19.7	0.7	-12.7	-17.4	0.9	-10.5	12.5	4.9	11.5	5.1	3.4	340	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	38.2	22.9	36.7	23.2	34.8	23.0	25.2	33.0	24.6	32.4	24.0	31.7	6.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.8	28.3	22.5	18.2	27.8	22.0	17.7	27.5	79.8	33.0	76.9	32.3	74.3	31.9	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	11.0	9.6	-19.8	40.7	2.4	1.7	-21.6	41.9	-23.0	42.9	-24.4	43.8	-26.1	45.0
			-20.2	26.6	2.3	1.1	-21.9	27.3	-23.2	27.9	-24.6	28.5	-26.2	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.1	2.0	7.8	13.2	18.9	24.5	27.0	26.0	21.5	14.2	7.0	0.9
	DBStd	10.81	5.92	6.23	5.99	5.31	4.66	3.80	3.46	3.36	4.53	5.30	5.43	5.61
	HDD10.0	1104	308	231	112	28	1	0	0	0	0	21	119	284
	HDD18.3	2621	564	457	328	168	53	3	0	0	20	147	340	540
	CDD10.0	2440	2	7	44	123	276	435	528	495	346	152	30	3
	CDD18.3	916	0	0	2	13	69	188	271	237	115	20	1	0
	CDH23.3	10588	0	2	33	174	728	2153	3309	2698	1246	234	12	0
	CDH26.7	5266	0	0	5	47	274	1060	1834	1416	558	71	1	0
Wind	WSAvg	4.4	4.2	4.4	5.0	5.4	4.7	4.8	4.1	3.7	4.1	4.4	4.4	4.2
Precipitation	PrecAvg	771	19	28	52	66	119	117	92	88	68	55	37	28
	PrecMax	1081	55	79	137	151	314	221	224	158	127	165	148	113
	PrecMin	445	2	0	0	22	14	28	16	1	16	0	2	2
	PrecStd	151	14	23	37	34	72	51	54	42	32	45	35	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.3	26.8	31.2	34.8	38.7	41.1	40.0	37.4	31.7	25.0	18.6
		MCWB	9.4	11.5	14.8	18.6	21.2	22.8	23.0	23.0	22.3	19.6	14.3	11.3
	2%	DB	15.7	18.3	23.5	27.6	32.1	36.1	38.8	37.7	34.0	28.4	22.1	15.3
		MCWB	7.8	10.3	14.5	17.2	20.6	23.3	23.0	22.9	22.2	18.1	13.4	9.1
	5%	DB	12.5	15.3	21.1	25.0	29.6	34.2	37.1	36.0	32.3	26.0	18.8	12.6
		MCWB	6.3	8.8	13.1	16.2	20.2	22.7	23.3	23.2	21.7	17.0	11.7	7.4
	10%	DB	9.6	12.3	17.9	22.6	27.5	32.5	35.0	33.7	30.0	23.0	16.3	10.0
		MCWB	5.0	7.4	11.9	15.1	19.6	22.3	23.4	22.8	20.6	16.2	11.2	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	14.3	18.7	20.5	23.8	25.7	26.4	25.8	24.6	22.2	16.5	14.4
		MCDB	16.6	17.9	24.4	26.6	30.2	34.0	32.8	33.3	31.9	27.0	21.6	16.7
	2%	WB	8.5	11.8	16.6	19.1	22.6	24.7	25.4	25.0	23.4	20.5	15.0	10.3
		MCDB	13.6	15.6	20.5	24.6	28.9	32.6	33.0	32.8	30.7	25.3	18.8	13.3
	5%	WB	6.9	9.6	14.9	17.7	21.5	23.9	24.7	24.2	22.6	19.0	13.4	7.9
		MCDB	12.1	15.5	18.9	23.1	27.5	31.6	32.6	31.9	29.7	23.7	17.7	12.0
	10%	WB	5.1	7.2	12.3	16.3	20.4	23.1	24.0	23.6	21.7	17.3	11.7	5.9
		MCDB	9.8	11.9	17.5	21.9	25.9	30.5	32.0	31.2	28.5	21.7	15.9	9.9
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.7	13.6	13.5	13.1	12.9	13.0	13.0	13.6	13.8	13.4	12.1
		MCDBR	17.8	17.9	18.1	16.7	15.4	14.4	15.6	15.9	15.2	16.8	17.7	16.4
		MCWBR	11.2	11.3	10.1	8.9	6.8	5.0	4.2	4.9	5.7	8.3	10.0	10.8
	5% WB	MCDBR	16.7	16.4	14.1	14.6	13.2	13.4	13.1	13.1	13.3	13.0	14.3	14.8
		MCWBR	10.9	11.0	9.1	9.0	6.8	5.6	4.9	5.1	5.5	7.5	9.2	10.6
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.327	0.361	0.389	0.407	0.416	0.415	0.372	0.335	0.295	0.278	
	taud	2.511	2.492	2.416	2.347	2.345	2.321	2.332	2.310	2.393	2.477	2.518	2.519	
	Ebn at Noon	912	947	936	917	892	872	862	855	876	880	887	891	
	Edh at Noon	77	89	107	122	125	128	125	125	109	90	76	71	
All-Sky Solar Radiation	RadAvg	2.50	3.25	4.29	5.39	6.13	6.92	6.78	6.07	5.02	3.69	2.74	2.17	
	RadStd	0.24	0.42	0.31	0.33	0.49	0.43	0.40	0.34	0.32	0.41	0.26	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page